Theory

1. Central Idea

In the presence of the external NdFeB magnet's field, the magnetic domains align parallel to the field to minimise energy, and since at the surface we expect field lines to be parallel to the surface, we obtain a magnetisation that is circumferential in direction. A tiny dipole in one ring will interact with some dipole in the other ring (in addition to interacting with some dipole in the same ring, but this interaction does not affect the dynamics). The idea here is now to sum up all these interactions and find a total potential of interaction between the two rings. We can then construct a Lagrangian using this potential and obtain the angles rotated by each ring as a function of time.

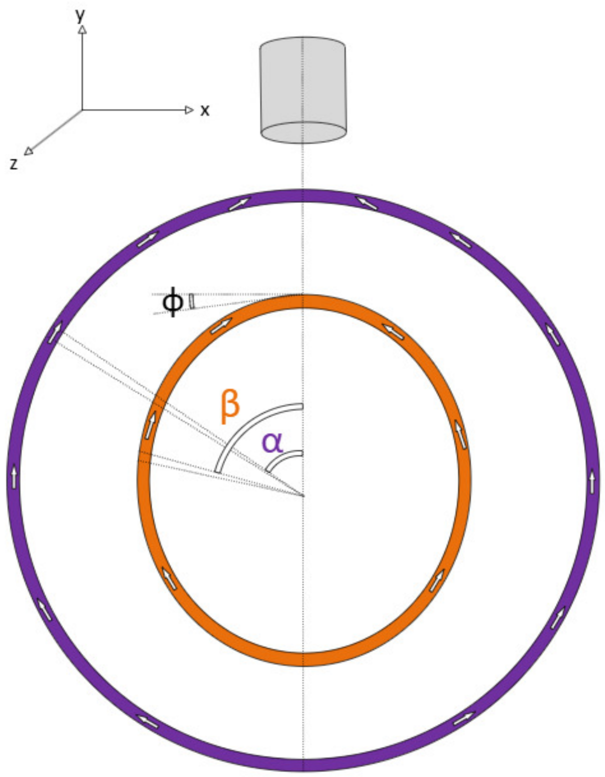
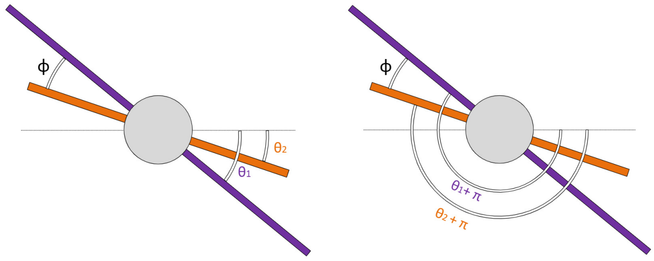


Figure 1: Schematic Diagram of the Rings

2. Obtaining The Magnetic Interaction

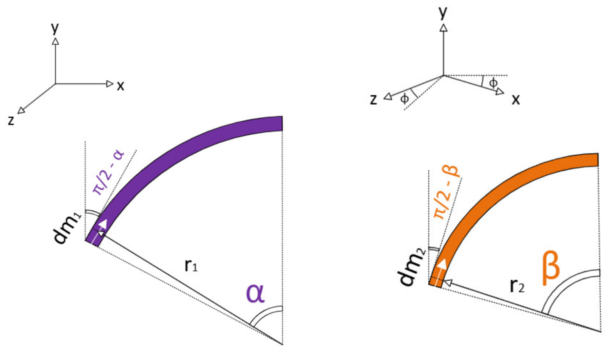
Let the angle rotated by the outer ring of radius R_1 be θ_1, that for the inner ring of radius R_2 be θ_2 and define $\phi := \theta_1 - \theta_2$ (see Figures 2 and 3(a)). Now consider one point dipole on the outer ring at some angle α and another on the inner ring at some angle β. The potential energy of interaction between two elemental point dipoles $d\vec{m}_1$ and $d\vec{m}_2$ with some separation vector $\vec{r}$ is given by [2]

$$dU = \frac{\mu_0}{4\pi r^5}(r^2(d\vec{m}_1 \cdot d\vec{m}_2) - 3(d\vec{m}_1 \cdot \vec{r})(d\vec{m}_2 \cdot \vec{r}))$$



(a) Top View (b) π symmetry

Figure 2



(a) (b)

Figure 3: Vectorial Diagrams

The position vector for the outer dipole, say $\vec{r_1}$, is given as

$$\vec{r_1} = -R_1 \sin\alpha\,\hat{x} + R_1 \cos\alpha\,\hat{y}$$

With magnitude dm_1 and pointing tangentially, we have

$$d\vec{m}_1 = dm_1 \cos\alpha\,\hat{x} + dm_1 \sin\alpha\,\hat{y}$$

Similarly, the position vector for the inner dipole $\vec{r_2}$ is given as

$$\vec{r_2} = -R_2 \sin\beta\cos\phi\,\hat{x} + R_2\cos\beta\,\hat{y} + R_2\sin\beta\sin\phi\,\hat{z}$$

And again,

$$d\vec{m}_2 = dm_2\cos\beta\cos\phi\,\hat{x} + dm_2\sin\beta\,\hat{y} - dm_2\cos\beta\sin\phi\,\hat{z}$$

The separation vector $\vec{r} := \vec{r_1} - \vec{r_2}$ is then

$$\vec{r} = (R_2\sin\beta\cos\phi - R_1\sin\alpha)\,\hat{x} + (R_1\cos\alpha - R_2\cos\beta)\,\hat{y}$$

$$-R_2\sin\beta\sin\phi\,\hat{z}$$

After some algebra, it can be shown that

$$r^2 = R_1^2 + R_2^2 - 2R_1R_2(\sin\alpha\sin\beta\cos\phi + \cos\alpha\cos\beta)$$

Now after evaluating the relevant dot products that appear in the potential, we obtain:

$$\vec{dm}_1 \cdot \vec{dm}_2 = dm_1 dm_2 (\cos\alpha\cos\phi\cos\beta + \sin\alpha\sin\beta)$$

$$\vec{dm}_1 \cdot \vec{r} = dm_1 R_2 (\cos\alpha\cos\phi\sin\beta - \sin\alpha\cos\beta)$$

$$\vec{dm}_2 \cdot \vec{r} = dm_2 R_1 (\cos\alpha\sin\beta - \sin\alpha\cos\beta\cos\phi)$$

Plugging these expressions into our expression for the potential and working through some algebra, we get

$$dU = \frac{\mu_0}{4\pi r^5} dm_1 dm_2 [(R_1^2 + R_2^2)(\cos\alpha\cos\beta\cos\phi + \sin\alpha\sin\beta) + R_1 R_2((\sin\alpha\sin\beta\cos\alpha\cos\beta)(1+\cos^2\phi) - \cos\phi\,(\sin^2\beta\cos^2\alpha + \sin^2\alpha\cos^2\beta + 2))]$$

By definition, we also have $dm = MdV$. Explicitly in this case,

$$dm_1 = M_s(\frac{\pi t^2}{4})(R_1 d\alpha)$$

$$dm_2 = M_s(\frac{\pi t^2}{4})(R_2 d\beta)$$

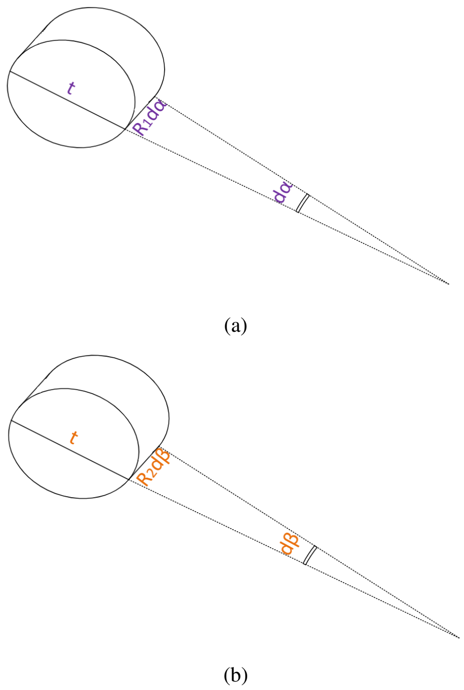


Figure 4: Volume Elements

Where, t is the thickness of the ring, and we make an assumption that the rings' magnetisations are in saturation, which is hence taken as M_s. So finally we have,

$$dU = \frac{\mu_0}{4\pi r^5}\frac{M_s^2\pi^2 t^4}{16} R_1 R_2 [(R_1^2+R_2^2)(\cos\alpha\cos\beta\cos\phi+\sin\alpha\sin\beta) + R_1 R_2((\sin\alpha\sin\beta\cos\alpha\cos\beta)(1+\cos^2\phi) - \cos\phi\,(\sin^2\beta\cos^2\alpha + \sin^2\alpha\cos^2\beta + 2))]\,d\beta\,d\alpha$$

After performing the double integral, the above U shall be a function of ϕ, the angular separation between the two rings. Note that we cannot naively integrate from 0 to 2π for α and β, as the direction of magnetisation reverses midway, so we need to be careful with the limits of integration.

Let the integrand be $f(\alpha,\beta)$. Then taking into account the direction of the dipoles,

$$\iint f(\alpha,\beta)\,d\beta\,d\alpha = \int_0^{\pi}\int_0^{\pi} f(\alpha,\beta)\,d\beta\,d\alpha + \int_0^{\pi}\int_{\pi}^{2\pi} f(\alpha,\beta)\,d(-\beta)\,d\alpha + \int_{\pi}^{2\pi}\int_0^{\pi} f(\alpha,\beta)\,d\beta\,d(-\alpha) + \int_{\pi}^{2\pi}\int_{\pi}^{2\pi} f(\alpha,\beta)\,d(-\beta)\,d(-\alpha)$$

Due to bilateral symmetry, the first and fourth integrals must be equal, and the second and third integrals must be equal. So,

$$U(\phi) = 2\int_0^{\pi}\int_0^{\pi} f(\alpha,\beta)\,d\beta\,d\alpha - 2\int_{\pi}^{2\pi}\int_0^{\pi} f(\alpha,\beta)\,d\beta\,d\alpha$$

Since the integrand is too complicated to integrate analytically, we resort to numerical methods, whose results will be discussed soon. We shall solve the double integral for some fixed ϕ, and then loop the code over small steps of $\delta\phi$ to obtain $U(\phi)$ in the range $[0, \pi]$. We expect $U(\phi)$ to be periodic with period π because a separation of ϕ and $\phi + \pi$ are equivalent (Figure 3(b)), so obtaining $U(\phi)$ in the range $[0, \pi]$ guarantees we have $U(\phi)\ \forall\ \phi$.

We split the domain $[0,\pi]$ for 100 equally sized steps for ϕ, and perform the integration using the double quadrature integration scheme to integrate under the suitable limits, and this integral is recorded for all 100 data points. We then obtain the following plot using the parameters given in Table 1.

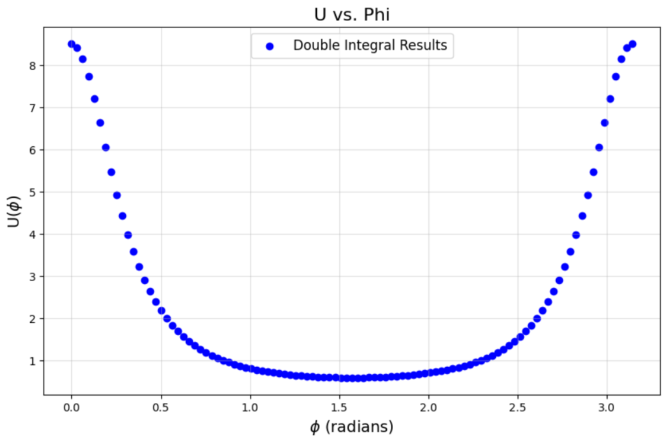


Figure 5: Interaction Potential

Parameter	Value
G (Shear modulus)	2 GPa to 10.7 GPa
r (Radius of string)	0.5 mm
l_1	20 cm
l_2	12.7 mm (0.5 inch)
k_1 (Torsional constant of l_1)	9.8×10^{-4} Nm to 5.25×10^{-3} Nm
k_2 (Torsional constant of l_2)	1.546×10^{-2} Nm to 8.27×10^{-2} Nm
M_s (Saturation magnetization)	1.7×10^6
M_1 (Mass of outer ring)	15.46 g
M_2 (Mass of inner ring)	11.73 g
R_1 (Radius of outer ring)	4 inches
R_2 (Radius of inner ring)	3 inches
Thickness of Set 1	3 mm
Thickness of Set 2	5 mm

Table 1: Parameters and their values.

3. Torsional Aspect of the Coupling

Since the rings are tied by strings that undergo torsion, we must take into account the energies stored in the strings due to this torsion. Due to the way we tie the strings, the equilibrium angle for no torsion may vary from setup to setup. It was observed that the inner string suffered no torsion when $\phi = \pi/2$. The lower string that is connected only to the outer ring undergoes a torsion of θ_1, and thus stores a potential energy of $\frac{1}{2}k_1\theta_1^2$. Similarly the upper string undergoes a torsion of $|\theta_1 - \theta_2 - \pi/2|$ and thus stores an energy of $\frac{1}{2}k_2(\theta_1 - \theta_2 - \pi/2)^2$, where each k is the respective torsional constant.

Also recall that $k = \frac{GJ}{L} = \frac{\pi r^4}{2L}$ for the string, where G is the shear modulus of the material, r is the radius, and L is the length of the string.

4. Solving for the Equations of Motion

Using the classical Lagrangian, we may directly write

$$\mathcal{L} = \frac{1}{2}I_1\dot{\theta_1}^2 + \frac{1}{2}I_2\dot{\theta_2}^2 - \frac{1}{2}k_1\theta_1^2 - \frac{1}{2}k_2(\theta_1 - \theta_2 - \pi/2)^2 - U(\theta_1 - \theta_2)$$

As per the Euler Lagrange Equations,

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} = \frac{\partial \mathcal{L}}{\partial \theta_i}$$

And so we get

$$\ddot{\theta_1} = \frac{1}{I_1}(-k_1\theta_1 - k_2(\theta_1 - \theta_2 - \pi/2) - \frac{\partial U}{\partial \theta_1})$$

$$\ddot{\theta_2} = \frac{1}{I_2}(k_2(\theta_1 - \theta_2 - \pi/2) - \frac{\partial U}{\partial \theta_2})$$

We use the above table of parameters as is in our setup. The value of M_s is derived from standard literature such as in [3].

5. Normal Modes of Small Oscillations

Using the chain rule, we can write the equations of motion in the following manner:

$$I_1\ddot{\theta_1} = -k_1\theta_1 - k_2(\theta_1 - \theta_2 - \pi/2) - \frac{\partial U}{\partial \phi}$$

$$I_2\ddot{\theta_2} = k_2(\theta_1 - \theta_2 - \pi/2) + \frac{\partial U}{\partial \phi}$$

We may expand $U(\phi)$ as a power series about $\pi/2$ and expect all odd powers to vanish because the function is even about $\pi/2$.

$$U = a_0 + a_1(\phi - \frac{\pi}{2})^2 + a_2(\phi - \frac{\pi}{2})^4 + a_3(\phi - \frac{\pi}{2})^6 + ...$$

$$\frac{\partial U}{\partial \phi} = 2a_1(\phi - \frac{\pi}{2}) + 4a_2(\phi - \frac{\pi}{2})^3 + ...$$

Define $a := 2a_1$ and neglect higher order variation,

$$I_1\ddot{\theta_1} + (k_1 + k_2 + a)\theta_1 - k_2(\theta_2 + a) - (k_2 + a)\frac{\pi}{2} = 0$$

$$I_2\ddot{\theta_2} - k_2(\theta_1 + a) + k_2(\theta_2 + a) + (k_2 + a)\frac{\pi}{2} = 0$$

This may be written as

$$\begin{bmatrix} I_1 & 0 \\ 0 & I_2 \end{bmatrix}\begin{bmatrix} \ddot{\theta_1} \\ \ddot{\theta_2} \end{bmatrix} + \begin{bmatrix} k_1 + k_2 + a & -(k_2 + a) \\ -(k_2 + a) & k_2 + a \end{bmatrix}\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} (k_2 + a)\frac{\pi}{2} \\ -(k_2 + a)\frac{\pi}{2} \end{bmatrix}.$$

Constant forcing terms will only change the equilibrium position, so we may just work with the following homogeneous equation.

$$\begin{bmatrix} I_1 & 0 \\ 0 & I_2 \end{bmatrix}\begin{bmatrix} \ddot{\theta_1} \\ \ddot{\theta_2} \end{bmatrix} + \begin{bmatrix} k_1 + k_2 + a & -(k_2 + a) \\ -(k_2 + a) & k_2 + a \end{bmatrix}\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Assume a solution

$$\begin{bmatrix}\theta_1\\ \theta_2\end{bmatrix} = \begin{bmatrix}A\\ B\end{bmatrix} e^{i\alpha t}$$

Then we get

$$\begin{bmatrix}k_1 + k_2 + a - \alpha^2 I_1 & -(k_2 + a)\\ -(k_2 + a) & k_2 + a - \alpha^2 I_2\end{bmatrix}\begin{bmatrix}A\\ B\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

Set determinant to 0 for non-trivial solutions:

$$k_1k_2 + k_2^2 + k_2a - \alpha^2 k_2 I_1 + k_1 a + k_2 a + a^2 - \alpha^2 a I_1 -$$

$$\alpha^2 k_1 I_2 - \alpha^2 k_2 I_2 - \alpha^2 a I_2 + \alpha^4 I_1 I_2 - k_2^2 - a^2 - 2k_2 a = 0$$

$$\alpha^4 I_1 I_2 - \alpha^2 (k_2 I_1 + a I_1 + k_1 I_2 + k_2 I_2 + a I_2) + k_1(k_2 + a) = 0$$

$$\alpha^4 I_1 I_2 - \alpha^2((I_1 + I_2)(k_2 + a) + k_1 I_2) + k_1(k_2 + a) = 0$$

$$\alpha^2 = \frac{k_1 I_2 + (I_1 + I_2)(k_2 + a) \pm \sqrt{\Delta}}{2 I_1 I_2}$$

$$\Delta = k_1^2 I_2^2 + (I_1 + I_2)^2 (k_2 + a)^2 + 2k_1 I_2 (k_2 + a)(I_2 - I_1)$$

Plugging in the values, we get

$$I_1 = \frac{1}{2} M_1 R_1^2 = 2 \cdot 10^{-5}\ ,\ \ I_2 = \frac{1}{2} M_2 R_2^2 = 0.855 \cdot 10^{-5}$$

a = 0.036 is obtained from computing the second derivative of U at $\pi/2$. Subsequently,

$$\begin{bmatrix}-0.137 & -0.059\\ -0.059 & -0.025\end{bmatrix}\begin{bmatrix}A\\ B\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

This gives the normal mode (1,2.33), with $\omega_1 = 99.40$ rad/s

$$\begin{bmatrix}0.060 & -0.059\\ -0.059 & 0.058\end{bmatrix}\begin{bmatrix}A\\ B\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

And this gives the normal mode (1,-1), with $\omega_2 = 8.36$ rad/s. So finally, for small oscillations we have

$$\begin{bmatrix}\theta_1\\ \theta_2\end{bmatrix} = c_1 \begin{bmatrix}1\\ 2.33\end{bmatrix} \cos(99.40t) + c_2 \begin{bmatrix}1\\ -1\end{bmatrix} cos(8.36t)$$

Observe how $(1, -1)$ being a normal mode, means exactly that $\phi = \theta_1 - \theta_2$ executes pure oscillations with time period $= \frac{2\pi}{\omega_2} = \frac{2\pi}{8.36} = 0.752$ s. This may be measured through experiment and be corresponded with the prediction.

It was also observed that the other normal mode is very nearly the ratio I_1/I_2 (Here $\frac{2}{0.855} = 2.33$). Unfortunately, this could not be analytically explained.

References

[1] Javier García et al. "Magnetization Reversal Process and Magnetostatic Interactions in Fe56Co44/SiO2/Fe3O4 Core/Shell Ferromagnetic Nanowires with Non-Magnetic Interlayer". In: *Nanomaterials* 11.9 (2021). ISSN: 2079-4991. DOI: 10.3390/nano11092282. URL: https://www.mdpi.com/2079-4991/11/9/2282.

[2] David J. Griffiths. *Introduction to Electrodynamics*. 4th ed. Cambridge University Press, 2017.

[3] D. Jiles. *Introduction to Magnetism and Magnetic Materials*. 3rd ed. CRC Press, 2015. DOI: https://doi.org/10.1201/b18948.

[4] David Laroze et al. "Characterization of the Chaotic Magnetic Particle Dynamics". In: *IEEE Transactions on Magnetics* 47.10 (2011), pp. 3032–3035. DOI: 10.1109/TMAG.2011.2158072.

[5] David Laroze et al. "Magnetization Dynamics Under a Quasiperiodic Magnetic Field". In: *IEEE Transactions on Magnetics* 48.11 (2012), pp. 3567–3570. DOI: 10.1109/TMAG.2012.2207378.

[6] Tadashi F. Tokieda. "The Hopping Hoop". In: *The American Mathematical Monthly* 104.2 (1997), pp. 152–154.